

C278

College Algebra

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A community-college tutor who has coached the math team for six years writes $-3x + 7 > 22$ on the board for a review session of 28 students. Which set describes every solution?

- A. $x < -5$
- B. $x \leq -5$
- C. $x < -29/3$
- D. $x > -5$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which inequality has the solution set $x \geq 4$?

- A. $2x \leq 8$
- B. $-2x \leq -8$
- C. $-2x \geq -8$
- D. $-2x \leq 8$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An analyst at a regional shipping firm that runs 40 trucks models a cost ratio with $f(x) = \sqrt{(x - 3) / (x - 7)}$. The model was approved by the finance team last quarter. What is the domain of f ?

- A. $x \geq 3$
- B. $x > 3$ and $x \neq 7$
- C. All real numbers except 3 and 7
- D. $x \geq 3$ and $x \neq 7$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A town of 50,000 people is growing 4% per year. The mayor, who won re-election last spring, asks the planning office for a 3-year projection before a new school bond vote. About how many residents should the office project?

- A. About 56,243
- B. 3,200,000
- C. 56,000
- D. About 44,237

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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5. [TRANSFER — reversed stem / different variable]

A car worth \$30,000 loses 15% of its value each year. Which model gives its value after t years?

- A. $V = 30,000(1.15)^t$
- B. $V = 30,000(0.85)^t$
- C. $V = 30,000 - 0.15t$
- D. $V = 30,000(0.15)^t$

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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6.

A lab technician who calibrates sensors on the night shift needs to solve $\log_2(x) + \log_2(x - 2) = 3$ to set a threshold. Her supervisor wants the answer before the 7 a.m. handoff. What is the solution?

- A. $x = 4$ or $x = -2$
- B. $x = 5$
- C. $x = 1 + \sqrt{7}$
- D. $x = 4$

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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7. [TRANSFER — reversed stem / different variable]

Which equation is equivalent to $\log_3(x) = 4$?

- A. $x = 4^3 = 64$
- B. $x = \pm 81$
- C. $x = 3^4 = 81$
- D. $x = 3 \times 4 = 12$

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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8.

A plumbing company founded in 1998 charges a flat trip fee plus an hourly rate. A 2-hour job costs \$190 and a 5-hour job costs \$385. The dispatcher notes most calls come in on Mondays. What is the trip fee?

- A. \$60
- B. \$65
- C. \$95
- D. \$77

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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9.

A coach at a high school with 1,200 students launches a ball whose height in feet is $h(t) = -16t^2 + 64t + 5$, where t is seconds. The launch happened on a windless afternoon. What is the ball's maximum height?

- A. 197 feet
- B. 5 feet
- C. 69 feet
- D. 2 feet

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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10. [TRANSFER — reversed stem / different variable]

A quadratic function $f(x) = ax^2 + bx + c$ has a minimum value. What must be true?

- A. $a < 0$
- B. $a = 0$
- C. $a > 0$ and $c > 0$
- D. $a > 0$

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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11.

A bakery owner who opened a second location in March writes two constraints for a pricing model: $2x + 3y = 12$ and $4x + 6y = 20$. Her accountant has a CPA license. How many solutions does the system have?

- A.** Exactly one, because every two-equation system has one
- B.** Exactly one, (3, 2)
- C.** No solution
- D.** Infinitely many solutions

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

An engineer at a firm with offices in three states models efficiency with $f(x) = (3x^2 + 1) / (x^2 - 4)$. The model has been used in two prior bids. What is the horizontal asymptote?

- A.** No horizontal asymptote
- B.** $y = 3$
- C.** $y = -1/4$
- D.** $y = 0$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A community-college tutor ~~who has coached the math team for six years~~ writes $-3x + 7 > 22$ on the board ~~for a review session of 28 students~~. Which set describes every solution?

Target word: solution set of the inequality **Decides:** Dividing by -3 reverses the sign.

A	Correct	$-3x > 15$, then dividing by -3 flips the sign: $x < -5$.
B	Adjacent category	Changes a strict inequality into an inclusive one; -5 itself makes the left side equal 22, not greater.
C	Inverted operation	Adds 7 to both sides instead of subtracting it.
D	Inverted operation	Divides by -3 but forgets to reverse the sign.

2. Answer B Concept 1 • transfer

Which inequality has the solution set $x \geq 4$?

Target word: reverse: find the inequality that produces $x \geq 4$ **Decides:** Dividing $-2x \leq -8$ by -2 flips $\leq$ to $\geq$.

A	Confusion set	Positive coefficient, no flip: $x \leq 4$.
B	Correct	Dividing by -2 reverses the sign: $x \geq 4$.
C	Inverted operation	Dividing by -2 flips this to $x \leq 4$.
D	Dropped element	Drops the sign on the constant: gives $x \geq -4$.

3. Answer D Concept 2

An analyst at a regional shipping firm ~~that runs 40 trucks~~ models a cost ratio with $f(x) = \sqrt{(x - 3) / (x - 7)}$. ~~The model was approved by the finance team last quarter.~~ What is the domain of f ?

Target word: domain **Decides:** Root needs $x - 3 \geq 0$; denominator needs $x \neq 7$.

A	Dropped element	Keeps the root restriction but drops the zero denominator at $x = 7$.
B	Adjacent category	$\sqrt{0} = 0$ is allowed, so $x = 3$ belongs in the domain.
C	Confusion set	Treats the root like a denominator; values below 3 give a negative radicand.
D	Correct	The radicand must be nonnegative and the denominator nonzero.

4. Answer A Concept 3

A town of 50,000 people is growing 4% per year. The mayor, ~~who won re-election last spring~~, asks the planning office for a 3-year projection ~~before a new school bond vote~~. About how many residents should the office project?

Target word: population after 3 years **Decides:** Compounding: $50,000 \times 1.04^3$.

A	Correct	$50,000 \times 1.04^3 = 50,000 \times 1.124864 \approx 56,243$.
B	Confusion set	Uses the rate 4 as the base instead of the growth factor 1.04.
C	Adjacent category	Adds 4% of the original three times (linear), ignoring compounding.
D	Inverted operation	Uses 0.96, a decay factor, for a growing town.

5. Answer B Concept 3 • transfer

A car worth \$30,000 loses 15% of its value each year. Which model gives its value after t years?

Target word: decay model **Decides:** Decay factor = $1 - 0.15 = 0.85$.

A	Inverted operation	That is 15% growth, not loss.
B	Correct	Each year keeps 85% of the prior value.
C	Adjacent category	A linear model subtracting a tiny flat amount; percent loss compounds.
D	Confusion set	Uses the rate lost as the factor kept.

6. Answer D Concept 4

A lab technician ~~who calibrates sensors on the night shift~~ needs to solve $\log_2(x) + \log_2(x - 2) = 3$ to set a threshold. ~~Her supervisor wants the answer before the 7 a.m. handoff.~~ What is the solution?

Target word: solution of the log equation **Decides:** Product rule then check the domain: $x(x - 2) = 8$.

A	Dropped element	Skips the domain check; $\log_2(-2)$ is undefined.
B	Confusion set	Treats the sum of logs as $\log(2x - 2)$, confusing it with the log of a sum.
C	Inverted operation	Multiplies $2 \times 3 = 6$ instead of raising $2^3 = 8$.
D	Correct	$x(x - 2) = 2^3 = 8$ gives $x = 4$ or $x = -2$; $x = -2$ makes the logs undefined.

7. Answer C Concept 4 • transfer

Which equation is equivalent to $\log_3(x) = 4$?

Target word: exponential form **Decides:** $\log_b(x) = y$ means $b^y = x$.

A	Inverted operation	Swaps base and exponent.
B	Adjacent category	Treats it like an even-power equation; log arguments must be positive.
C	Correct	The base raised to the log's value gives the argument.
D	Confusion set	Treats the log as multiplication.

8. Answer A Concept 5

A plumbing company ~~founded in 1998~~ charges a flat trip fee plus an hourly rate. A 2-hour job costs \$190 and a 5-hour job costs \$385. ~~The dispatcher notes most calls come in on Mondays.~~ What is the trip fee?

Target word: intercept (flat fee) **Decides:** Slope = $(385 - 190)/3 = \$65/\text{hr}$; fee = $190 - 2(65)$.

A	Correct	Rate = \$65 per hour; $190 - 130 = \$60$.
B	Confusion set	That is the hourly rate (slope), not the fee (intercept).
C	Dropped element	$190 \div 2$ ignores the flat fee and treats the whole bill as hourly.
D	Wrong stem number	$385 \div 5$ uses one job's total as if it were all hourly.

9. Answer C Concept 6

A coach ~~at a high school with 1,200 students~~ launches a ball whose height in feet is $h(t) = -16t^2 + 64t + 5$, where t is seconds. ~~The launch happened on a windless afternoon.~~ What is the ball's maximum height?

Target word: maximum value (height), not time **Decides:** Vertex at $t = -64/(2 \cdot -16) = 2$; $h(2) = 69$.

A	Inverted operation	Adds the squared term instead of subtracting: $64 + 128 + 5$.
B	Wrong stem number	The starting height, $h(0)$.
C	Correct	$h(2) = -64 + 128 + 5 = 69$.
D	Confusion set	That is the time of the maximum, not the height.

10. Answer D Concept 6 · transfer

A quadratic function $f(x) = ax^2 + bx + c$ has a minimum value. What must be true?

Target word: reverse: from min to the sign of a **Decides:** A minimum means the parabola opens up.

A	Inverted operation	A negative a opens down and gives a maximum.
B	Confusion set	Then the function is linear and has no minimum.
C	Adjacent category	The constant term only shifts the graph; it does not decide min vs. max.
D	Correct	Opening upward requires a positive leading coefficient.

11. Answer C Concept 7

A bakery owner ~~who opened a second location in March~~ writes two constraints for a pricing model: $2x + 3y = 12$ and $4x + 6y = 20$. ~~Her accountant has a CPA license.~~ How many solutions does the system have?

Target word: number of solutions **Decides:** Left sides are proportional ($\times 2$) but $12 \times 2 \neq 20$: parallel lines.

A	Absolute language	Systems can have zero, one, or infinitely many solutions.
B	Dropped element	Checks only the first equation; $4(3) + 6(2) = 24 \neq 20$.
C	Correct	Same slope, different intercepts: the lines never meet.
D	Confusion set	That needs the constants to scale too (24, not 20).

12. Answer B Concept 8

An engineer ~~at a firm with offices in three states~~ models efficiency with $f(x) = (3x^2 + 1) / (x^2 - 4)$. ~~The model has been used in two prior bids.~~ What is the horizontal asymptote?

Target word: horizontal asymptote **Decides:** Equal degrees: ratio of leading coefficients $3/1$.

A	Adjacent category	That applies when the numerator's degree is higher.
B	Correct	Degrees are equal, so the asymptote is $3/1$.
C	Wrong stem number	Divides the constant terms instead of the leading coefficients.
D	Confusion set	That rule applies only when the numerator's degree is lower.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Inequalities flip under negatives	Multiplying or dividing both sides of an inequality by a negative number reverses the inequality sign.	Solving an inequality exactly like an equation and keeping the sign.	A negative coefficient on the variable.
2	Domain of a function	Exclude inputs that make a denominator zero or put a negative under an even root.	Excluding only one restriction when both a root and a denominator appear.	"Which values can x take?" with a fraction or square root.
3	Exponential growth and decay	$A(t) = P(1 + r)^t$ for growth and $P(1 - r)^t$ for decay; the growth factor is $1 + r$, not r .	Using the rate itself as the base, or growing by a flat amount each period.	"Percent per year" compounding over several periods.
4	Logarithms undo exponents	$\log_b(x) = y$ means $b^y = x$; log rules turn a sum of logs into the log of a product, and answers must keep every log argument positive.	Writing $\log(a) + \log(b) = \log(a + b)$, or keeping an answer that makes a log argument negative.	A log equation with more than one log term.
5	Slope and intercept in linear models	In a linear model, slope is the rate of change per unit and the intercept is the starting (fixed) value.	Dividing a total by the input and calling it the rate.	Two data points and a "flat fee plus per-unit" setup.
6	Quadratic vertex	For $f(x) = ax^2 + bx + c$ the vertex is at $x = -b/(2a)$; $a > 0$ opens up (minimum), $a < 0$ opens down (maximum).	Reporting the x-value of the vertex when the question asks for the maximum value.	"Maximum" or "minimum" with a squared term.
7	Number of solutions of a linear system	Same slope and different intercepts means parallel lines and no solution; same line means infinitely many; different slopes means exactly one.	Assuming every two-equation system has exactly one solution.	One equation's coefficients are a multiple of the other's but the constants are not.
8	Asymptotes of rational functions	Vertical asymptotes come from zeros of the reduced denominator; the horizontal asymptote comes from comparing degrees (equal degrees gives the ratio of leading coefficients).	Using the ratio of constant terms, or defaulting to $y = 0$.	Numerator and denominator of the same degree.

Course scope (public description): This course provides further application and analysis of algebraic concepts and functions through mathematical modeling of real-world situations. Topics include: real numbers, algebraic expressions, equations and inequalities, graphs and functions, polynomial and rational functions, exponential and logarithmic functions, and systems of linear equations.